25901112

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBTG200 - BIOSAFETY AND BIOETHICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

SCHEME

1. Which BSL requires a full-body, air-supplied positive pressure suit? [K] / [CO3]
   BSL 4 – 2 marks
2. List any two types of risks faced by an organization. [K] / [CO4]
   Any two types- 2 marks
3. What is informed consent? [U] / [CO5]
   Definition – 2 marks
4. What is "Analgesia"? [K] / [CO5]
   Analgesia –use of methods or drugs to relieve pain in laboratory animals- 2 marks
5. Indicate the process used to prevent disease spread. [U] / [CO1]
   Any two- 2 marks
6. Mention one function of ethics committees in biological research. [U] / [CO1]
   Any one function- 2 marks
7. Name the organization responsible for issuing the Declaration of Helsinki. [U] / [CO5]
   World Medical Association– 2 marks
8. What is the term for the voluntary agreement given by a minor to participate in research? [U] / [CO5]
   Assent- 2 marks
9. Identify the equipment used for sterile airflow in Laboratory. [U] / [CO2]
   LAF- 2 marks
10. Identify any two types of hand protection. [U] / [CO3]

Any two- 2 marks

11. Which ethical term refers to data protection? [U] / [CO5]
    Confidentiality-2 marks
12. Who has the power to shut down an unethical study? [U] / [CO5]
    Institutional Ethics committee (IEC) or Institutional Review Board (IRB) or Ethics Committee.- 2 marks

[2x10 = 20]

Part B-Short Essay-Answer any 4 questions

Each question carries 5 marks

13. Apply ethical principles to propose a strategy that minimizes the use of animals in research.[A] / [CO5]
    Minimum 5 points-5 marks
14. Describe the difference between institutional review board and ethic committees. [U] / [CO5]
    Minimum 5 points-5 marks
15. Discuss five major types of hazards associated with laboratory or industrial workplaces. [An] / [CO4]
    Five types- 5 marks
16. Explain the purpose of the Cartagena Protocol on Biosafety. [U] / [CO3]
    Cartagena Protocol-Minimum points- 5 marks
17. Explain the role of communication systems during biosafety incidents. [K] / [CO1]
    Minimum 5 Points- 5 marks
18. Explain different physical containment measures. [U] / [CO4]
    Explain- minimum 5 types- 5 marks

[5x4 = 20]

Part C-Essay type questions-Answer any 2 questions

Each question carries 15 marks

19. Explain how experimental plant research can contribute to sustainable agriculture.[E] / [CO5]
    Plant research- introduction- 5 marks, Role of plant research in sustainable agriculture- 5 points- 10 marks
20. Identify the regulatory guidelines governing rDNA research in India and state their significance.[K] / [CO2]
    R DNA research-Introduction- 2 marks, guidelines-8 marks, significance- 5 marks.
21. Describe in detail the SOP for aseptic techniques and microbial culture handling in the laboratory.[U] / [CO2]
    SOP- introduction- 3 marks, SOP for aseptic techniques- 5 marks, SOP for microbial culture handling (points)- 7 marks
22. Explain the major components of a biosafety cabinet and their functions. [K] / [CO3]
    Biosafety-Introduction- 3 marks, biosafety cabinet – components- 4 marks, functions- 8 marks

[15x2 = 30]